X-ray physics

What makes an X-ray image useful?

A detector measures the X-rays that reach it after they pass through the body.

- **Physical signal:** transmitted X-ray photons.
- **Detector measurement:** a signal at each detector pixel.
- **Contrast mechanism:** different paths transmit different fractions of the incident beam.

Chest radiography

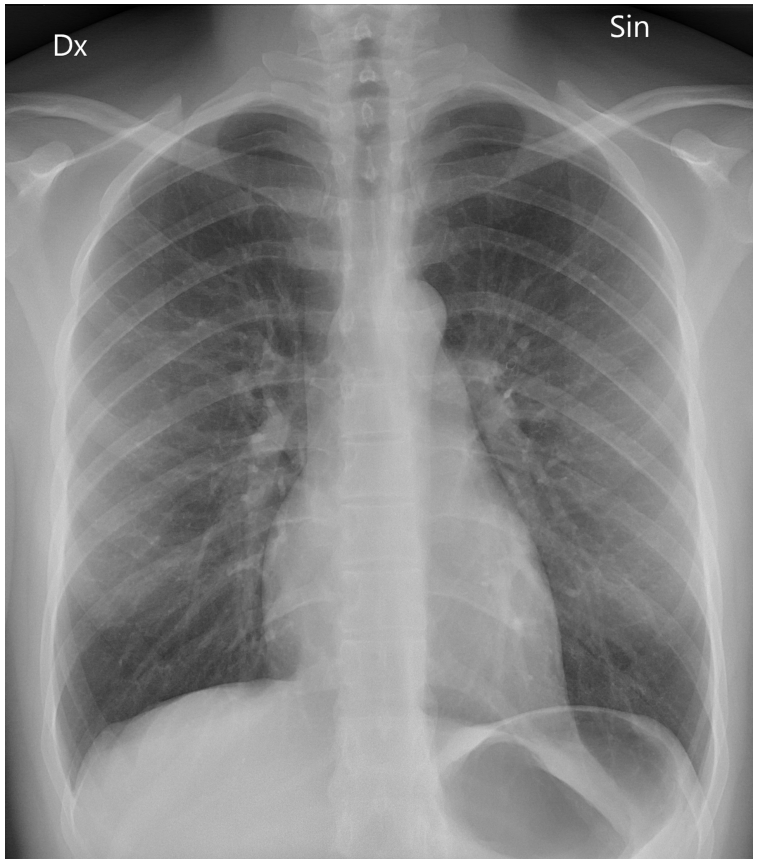


Figure 1: Normal chest radiograph

Air-filled lungs generally transmit more X-rays than paths through the heart and nearby soft tissue.

An x-ray can show changes in the lung pattern or help check the position of a tube or line.

Mammography reveals small differences

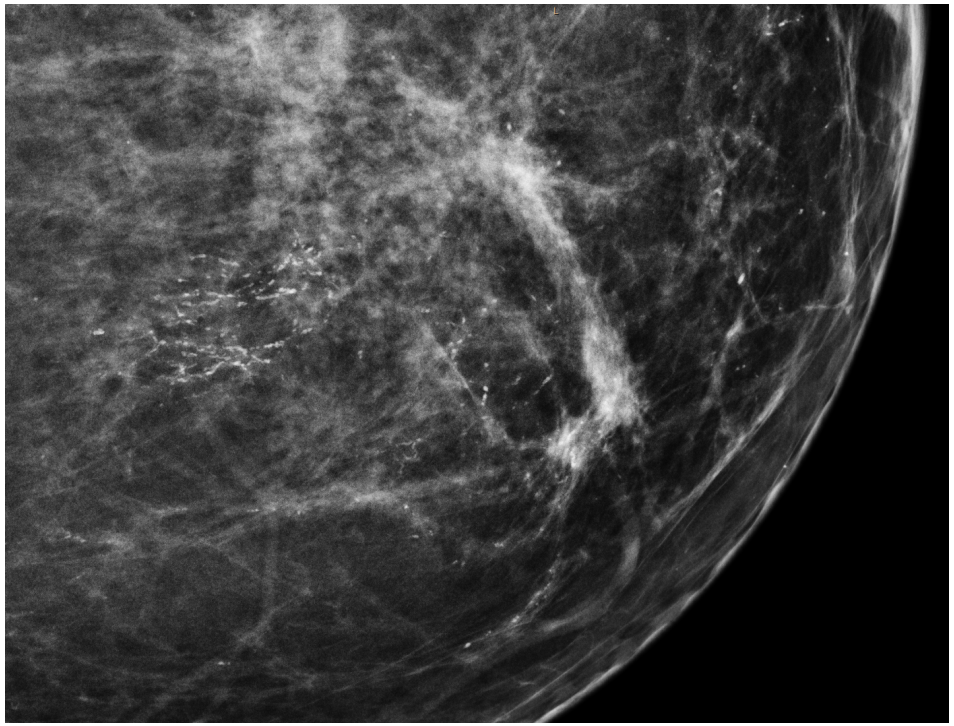


Figure 2: Mammogram detail showing microcalcifications

Mammography uses a lower-energy spectrum than general radiography to help distinguish similar breast tissues. Tiny calcium deposits stand out because they attenuate more strongly than nearby tissue. Small features require good spatial resolution.

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Fluoroscopy shows motion during a procedure

Fluoroscopy displays a sequence of X-ray projections during an exam or procedure. Named because historically the detector was a fluorescent screen.

- A clinician can follow a catheter as it moves.
- Iodine contrast can increase attenuation in blood vessels, making their paths easier to see.
- The engineering task is to show changes quickly for guidance.

How fine a pattern can we see?

The **pixel pitch** p is the center-to-center spacing between pixels. A bright bar and a dark bar make one **line pair**, or one cycle.

Detector examples	Pixel pitch p	Sampling limit $1/(2p)$
Chest radiography	0.14–0.20 mm	about 2.5–3.6 line pairs/mm
Digital mammography	0.07–0.10 mm	about 5–7 line pairs/mm

These are sampling limits. Blur reduces the contrast of fine patterns, and noise can hide them.

Questions: A pattern has 4 line pairs/mm at the detector. Which of these detectors sample it finely enough? What could still make the bars hard to see?

X-ray tube

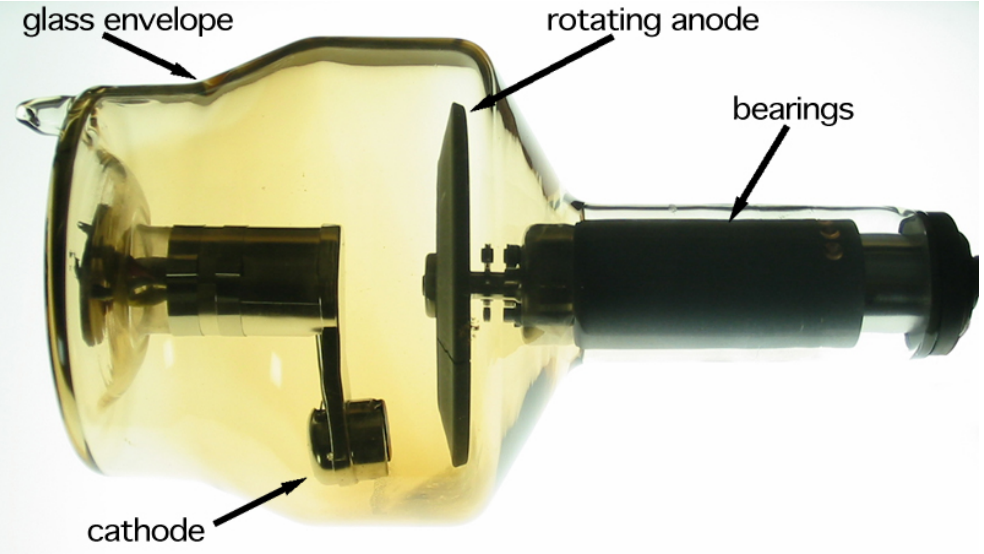


Figure 3: Labeled rotating-anode X-ray tube insert

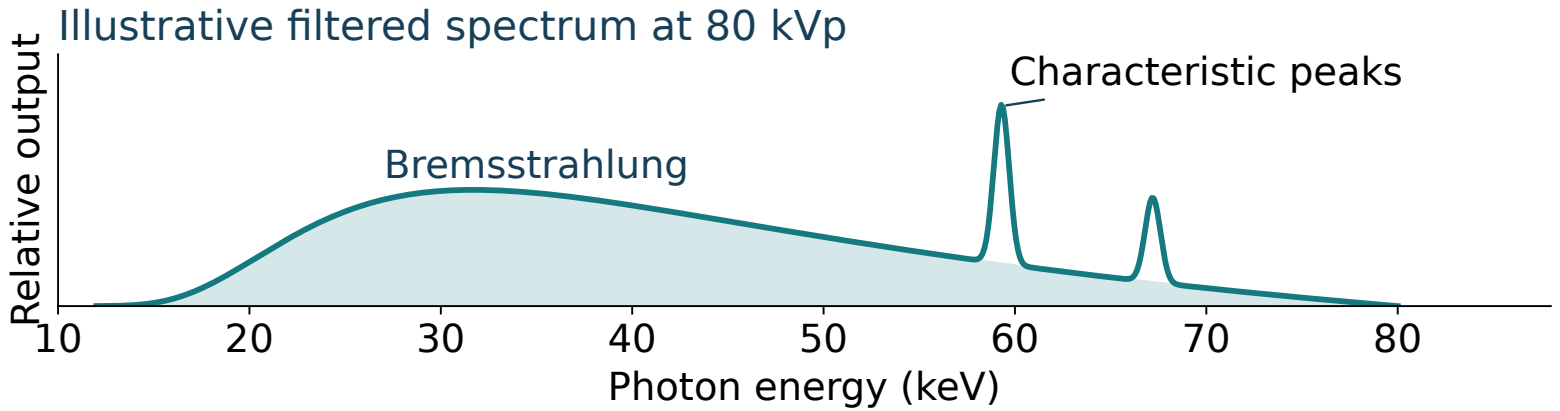
A heated **cathode** supplies electrons. A voltage accelerates them toward the **anode**, the target. The anode is spun to spread out the heat and prevent it from melting!

When the electrons interact with the target, some of their energy becomes X-rays. Most becomes heat.

The tube produces a range of photon energies

Bremsstrahlung is the broad spectrum emitted as electrons are deflected and slow in the target.

Characteristic X-rays appear at discrete energies set by the target material.



Tube voltage and current

- **Tube voltage** changes the electron energy and the photon spectrum. An electron accelerated through 80 kV gains 80 keV. That is the upper x-ray photon energy at that tube voltage.
- **Tube current** At fixed tube voltage and filtration, more current primarily means more photons. Current multipled by time, mAs, relates to the total number of photons.

Example: two settings for the same tube

Both exposures use 80 kVp and the same filtration.

Exposure	Tube current	Exposure time
A	100 mA	40 ms
B	200 mA	20 ms

1. Calculate the mAs for each. Which produces more X-ray photons?
2. Which has the higher maximum photon energy?
3. If the object is moving, which exposure would give less motion blur?

Photons interaction: pass through, scatter, or absorbed

- **Transmission:** the photon continues along the original path.
- **Photoelectric absorption:** the photon transfers its energy in an interaction that ejects a bound electron.
- **Compton scattering:** the photon transfers some energy to an electron and changes direction.

Attenuation is the loss of photons from the primary beam. Some scattered photons still reach the detector.

Transmission versus depth

For a single photon energy, the transmission, T, is exponential in depth.

$$T = \frac{I}{I_0} = e^{-\mu x}.$$

- I_0 and I are the incident and transmitted signals.
- x is the path length in the material, in mm.
- μ is the linear attenuation coefficient at this energy of the material, in mm^{-1}. Depends on electron density.

Here's a web tool for linear attenuation coefficient

μx **is dimensionless.**

What fraction is transmitted?

Suppose $\mu = 0.025$ mm^{-1}.

Path length	Attenuation exponent μx	Transmitted fraction
20 mm	0.50	$e^{-0.50} \approx 0.61$
40 mm	1.00	$e^{-1.00} \approx 0.37$

Why does doubling the thickness square the transmitted fraction?

The linear attenuation coefficient μ has units of inverse length. Examples are mm^{-1} or cm^{-1}. Length and inverse length must have matching units so that μx is dimensionless. The product μx must stay the same indendent of units. So, $5\text{mm}^{-1} = 50\text{cm}^{-1}$. A path length of 1 mm $= 0.1$ cm gives $\mu x = 5$ with both.

A ray can cross more than one material

For two layers, transmission fractions multiply:

$$\frac{I}{I_0} = e^{-\mu_1 x_1} e^{-\mu_2 x_2}.$$

Taking the negative logarithm makes the contributions add:

$$-\ln(I/I_0) = \mu_1 x_1 + \mu_2 x_2.$$

Discuss: What if the two layers trade places?

The spectrum of a beam changes as it passes through tissue

Lower-energy photons are often attenuated more strongly than higher-energy photons.

The transmitted beam then has fewer photons and a higher average photon energy. This is **beam hardening**.

A single value of the attenuation coefficient is a useful starting model. For a broad beam, transmission depends on the spectrum and on how attenuation changes with energy.

What can one measurement tell us?

Compare two paths at the same photon energy:

- Path A: 20 mm at $\mu = 0.025$ mm^{-1}.
- Path B: 10 mm at $\mu = 0.050$ mm^{-1}.

Which path transmits more? What could you infer if you measured only the transmitted fraction?

Absorbed dose is energy deposited per mass

To find the **average absorbed dose** in a region of tissue, divide the energy deposited in that region by its mass.

$$D = \frac{E_{\text{deposited}}}{m}, \qquad 1\ \text{Gy} = 1\ \text{J/kg}.$$

A photon can scatter out of the primary beam and carry away some energy. To calculate dose, we need the energy deposited in the tissue.

Dose estimates use measurements of the X-ray output and a model of the tissue and beam geometry.

Example: estimate the dose to a region of tissue

Suppose 2.0×10^9 photons of 50 keV each enter a region of tissue with a mass of 0.10 kg. The region absorbs 25% of the **total x-ray energy**.

1. Find the energy deposited in the region, in joules.
2. Find the average absorbed dose in Gy and in mGy.
3. At the same energy and geometry, what happens to the dose if the number of incident photons doubles?

Use 1 keV $= 1.602 \times 10^{-16}$ J.

Reading and image sources

IAEA, *Diagnostic Radiology Physics*, chapters 5 and 6.

- FDA overview of medical X-ray imaging
- FDA mammography and fluoroscopy
- NIST X-ray attenuation data
- More NIST X-ray attenuation data
- Representative detector pitches for chest radiography and mammography
- IAEA, Dosimetry in Diagnostic Radiology